3.7 Tangent Planes and Normal Vectors_Guide

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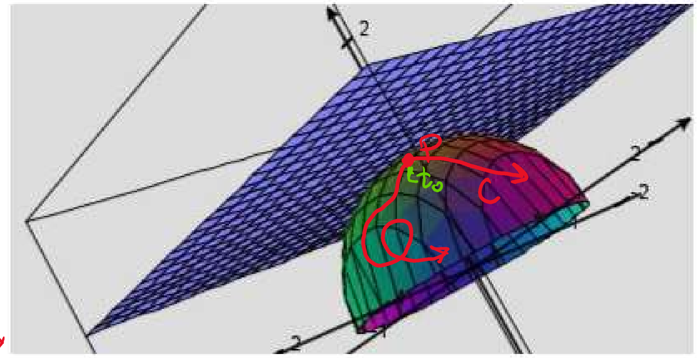
3.7 Tangent Planes an...

3.7 Tangent Planes and Normal Vectors

In this section we are interested in how to find the equation of a tangent plane to a surface at some point on the surface.

Motivation: Suppose that σ is a surface with equation $F(x, y, z) = k$ where k is a constant and F has continuous first partial derivatives. Let C be an arbitrary smooth curve that lies on σ and passes through the point P: (x_0, y_0, z_0).



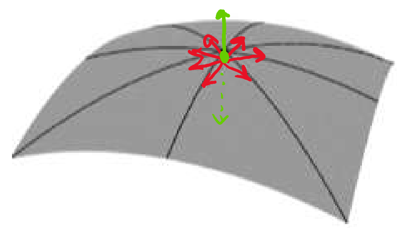
From 2.1 C would have some formula $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$

Let t_0 be the t-value such that $\vec{r}(t_0) = \langle x_0, y_0, z_0 \rangle$

Since $\vec{r}(t)$ lies on σ we know that any point of the form $(x(t), y(t), z(t))$

$$F(x(t), y(t), z(t)) = k$$

$$\begin{array}{ccc} & F & \\ x & y & z \\ | & | & | \\ t & t & t \end{array}$$



$$\frac{\partial F}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial F}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial F}{\partial z} \cdot \frac{dz}{dt} = 0$$

$$\left\langle \frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \right\rangle \cdot \left\langle \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right\rangle = 0$$

$$\vec{\nabla} F \cdot \vec{r}\,'(t) = 0$$

Since $\vec{\nabla} F$ has to be perpendicular to every $\vec{r}\,'(t_0)$ then it must be parallel to the binormal vector.

So $\vec{\nabla} F$ can act as a normal vector for the plane!

Definition: The tangent plane to the level surface $F(x, y, z) = k$ at the point P: (x_0, y_0, z_0) (assuming F has continuous first-order partials) is the plane given by the equation:

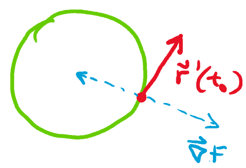
$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0$$

Note: In order for the above definition to work we need the gradient to be nonzero at our point of interest. We will say there is no tangent plane when the gradient is the zero vector at our point of interest.

Question: How would the above results change if we were dealing with level curves of the form $\underbrace{F(x,y)=k}_{x^2+y^2=1}$?

$\vec{\nabla} F = \langle F_x, F_y \rangle$ would be parallel to one of the normal vectors to the curve

b/c $\vec{\nabla} F \cdot \vec{r}'(t) = 0$

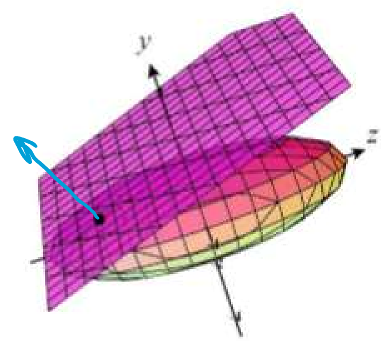


Example 1: Find an equation of the tangent plane at the point $(-2,1,-3)$ to the ellipsoid given by $\frac{x^2}{4}+y^2+\frac{z^2}{9}=3$. Note $F(x,y,z) = \frac{x^2}{4}+y^2+\frac{z^2}{9}$

Point: $(-2,1,-3)$

Normal vector $\vec{n}$: $\vec{\nabla} F = \langle \frac{1}{2}x, 2y, \frac{2}{9}z \rangle$

So $\vec{n} = \vec{\nabla} F(-2,1,-3) = \langle -1, 2, \frac{-2}{3} \rangle$



$$-1(x+2)+2(y-1)-\frac{2}{3}(z+3)=0$$

Motivation: Suppose we wish to find a tangent plane to a surface of the form $z=f(x,y)$.

Note that $z=f(x,y)$ is not in the form of $F(x,y,z)=k$, so we can't use our prior knowledge ... yet.

We move everything to one side of the equation.

$$z=f(x,y) \longrightarrow \underbrace{z-f(x,y)}_{F(x,y,z)} = \underbrace{0}_{k}$$

So we calculate $\vec{\nabla} F = \langle -f_x(x,y), -f_y(x,y), 1 \rangle = \vec{n}$

So the tangent plane at (x_0, y_0, z_0) is

$$-f_x(x_0,y_0)(x-x_0) - f_y(x_0,y_0)(y-y_0) + 1(z-z_0)=0$$

Theorem: Suppose f has continuous partial derivatives. An equation of the tangent plane to the surface $z = f(x,y)$ at the point P: (x_0, y_0, z_0) is:

$$z - z_0 = f_x(x_0,y_0)(x - x_0) + f_y(x_0,y_0)(y - y_0).$$

Proof: The proof of this theorem is shown in the motivation above.

Note: This definition agrees with the result we obtained at the beginning of this section, however the previous definition was more general. In fact, for most purposes memorizing the above formula is not necessary as we can treat all surfaces of the form $z = f(x,y)$ like they are surfaces of the form $F(x,y,z) = k$ like we did in the previous derivation and example. We note that this is how we actually construct the remaining tangent planes from this section.

Example 2: Find an equation of the tangent plane to surface determined by $f(x,y) = xe^{xy}$ at the point $(2,0,2)$

Note we have $z = xe^{xy} \longrightarrow \underbrace{z - xe^{xy}}_{F(x,y,z)} = \underbrace{0}_{k}$

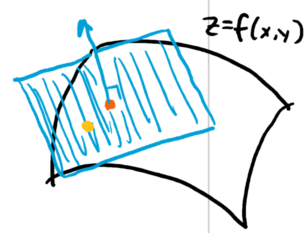
We calculate $\vec{\nabla} F = \langle -xe^{xy}y + (-1)e^{xy}, -x^2 e^{xy}, 1 \rangle$

$\vec{n} = \vec{\nabla} F(2,0,2) = \langle -1, -4, 1 \rangle$

The tangent plane is:

$$-1(x-2) - 4(y-0) + 1(z-2) = 0$$

Example 3: Find an equation of the tangent plane to the surface determined by $f(x,y) = \ln(x-2y)$ at the point $(3,1,0)$.



Note we have $z = \ln(x-2y) \longrightarrow \underbrace{z - \ln(x-2y)}_{F(x,y,z)} = 0$

We calculate $\vec{\nabla} F = \langle -\frac{1}{x-2y}, -\frac{1}{x-2y}(-2), 1 \rangle$

So $\vec{\nabla} F(3,1,0) = \langle -1, 2, 1 \rangle$

So the equation of the plane is:

$$-1(x-3) + 2(y-1) + 1(z-0) = 0$$

Note: $z = (x-3) - 2(y-1) = x - 2y - 1 \longrightarrow z = x - 2y - 1$

OR

$L(x,y) = x - 2y - 1$

Example 4: Use the tangent plane constructed in Example 3 to estimate the value of $f(x,y)$ at $(3.2, 1)$

Example 4: Use the tangent plane constructed in Example 3 to estimate the value of $f(x,y)$ at $(3.2,1)$ without the use of a calculator.

Note that $f(x,y) \approx L(x,y)$ as long as (x,y) is near $(3,1)$

So $f(3.2,1) \approx L(3.2,1) = (3.2) - 2(1) - 1 = 0.2$

Note: The notion of a tangent plane approximating a surface at a point is similar to how we used tangent lines to approximate curves in Calculus I. However, we can also extend ideas of Taylor series expansions from Calculus II (which allowed us to continue to better approximate curves in 2-spaces) to multivariate Taylor expansions which would allow us to continue to better approximate surfaces in 3-space!